

O M S B

Oman Medical Specialty Board

Evidence Based Medicine Course

Workshop Handout Kit

Date: 21 June 2026

Facilitators

Dr. Abdullah Al Alawi

Dr. Mohammed Al Rawahi

EBM Workshop 2026 — OMSB

Workshop Schedule

21 June 2026 · OMSB

Time	Activity / Session	Facilitator
7:30 – 8:00	Registration & Welcome Coffee	—
8:00 – 8:10	Introduction & Objectives	Dr. Al Alawi
8:10 – 8:40	Session 1 – PICO & Literature Searching	Dr. Al Alawi
8:40 – 9:00	Session 2 – Study Designs, Levels & Grades of Evidence	Dr. Al Alawi
9:00 – 9:20	Session 3a – Critical Appraisal of RCTs (Theory)	Dr. Al Alawi
9:20 – 9:45	Session 3b – Hands-on: RESTORE Trial	Dr. Al Alawi
9:45 – 10:00	Buffer / Q&A	—
10:00 – 10:30	<i>Coffee Break</i>	
10:30 – 11:10	Session 4 – Diagnostic Accuracy & Prognostic Studies	Dr. Al Rawahi
11:10 – 11:45	Session 5 – Hands-on: Observational Study Appraisal	Dr. Al Rawahi
11:45 – 12:45	Session 6 – AI in Medicine	Dr. Al Rawahi
12:45 – 1:20	<i>Prayer & Lunch</i>	
1:20 – 2:15	Session 7 – Systematic Reviews & Meta-Analysis	Dr. Al Alawi
2:15 – 2:30	Wrap-Up & Reflection	All

Learning Objectives

By the end of this workshop, participants will be able to:

1. Formulate focused clinical questions using the PICO framework
2. Identify study designs and classify levels and grades of evidence
3. Critically appraise RCTs, observational studies, and systematic reviews
4. Interpret forest plots, 2×2 tables, and key statistical measures
5. Conduct efficient PubMed literature searches using MeSH and Boolean operators
6. Recognize opportunities and limitations of AI tools in clinical practice

Note: All handout materials in this kit are designed for active use during sessions. Please bring a pen.

Evidence Based Medicine Course — Pre / Post Assessment

Name: _____

Level: _____

Date: _____

Circle the single best answer for each question.

Q1. A 68-year-old man with heart failure with reduced ejection fraction (HFrEF), already on guideline-directed medical therapy, is being considered for dapagliflozin. You want to search the literature for evidence. Which of the following is the best-formulated PICO question?

- A. Does dapagliflozin work in heart failure?
- B. In adults with HFrEF on standard therapy (P), does adding dapagliflozin (I) compared with placebo (C) reduce cardiovascular death or heart failure hospitalization (O)?
- C. In hospitalized patients (P), is dapagliflozin (I) better than other SGLT2 inhibitors (C) for mortality (O)?
- D. Should we use dapagliflozin for cardiac patients?

Q2. Researchers identified 150 patients with hepatocellular carcinoma and 300 patients without, then looked back at their prior hepatitis B status and aflatoxin exposure. What type of study design is this?

- A. Prospective cohort study
- B. Randomized controlled trial
- C. Case-control study
- D. Cross-sectional study

Q3. A new high-sensitivity troponin assay has a sensitivity of 95% and a specificity of 80% for acute myocardial infarction. In a patient presenting to the ED with chest pain, what does the high sensitivity of this test mean?

- A. A positive result confirms the diagnosis of MI
- B. A negative result makes MI very unlikely
- C. The test has a low false-positive rate
- D. The test is useful only for ruling in MI

Q4. A randomized controlled trial enrolled 1,000 patients with sepsis. In the treatment group, 15% of patients were lost to follow-up, compared with 3% in the control group. The investigators analyzed only patients who completed the study. Which of the following is the most important concern about this trial's validity?

- A. The sample size was too small
- B. Differential loss to follow-up and lack of intention-to-treat analysis may introduce bias
- C. The study was not double-blinded
- D. The primary outcome was poorly defined

Q5. In a meta-analysis forest plot, one of the included studies shows a 95% confidence interval that crosses the line of no effect (relative risk = 1.0). What is the correct interpretation?

- A. The study shows a statistically significant benefit of the intervention
- B. The study result is not statistically significant at the 5% level
- C. The study should be excluded from the meta-analysis
- D. The pooled estimate is unreliable

PICO Worksheet

PICO is a framework for formulating focused clinical questions: **P**opulation/**P**atient, **I**ntervention, **C**omparison, **O**utcome. A well-built PICO question guides your literature search and helps identify the best available evidence.

Vignette 1

A 65-year-old man presents with newly diagnosed atrial fibrillation, CHA₂DS₂-VASc score of 3. You are considering direct oral anticoagulants (DOACs) versus warfarin for stroke prevention.

P	
I	
C	
O	
Clinical Question	

Vignette 2

A 55-year-old woman with type 2 diabetes and CKD stage 3 (eGFR 42). You want to know if adding an SGLT2 inhibitor will slow kidney disease progression.

P	
I	
C	
O	
Clinical Question	

Vignette 3

A 70-year-old man admitted with community-acquired pneumonia requiring oxygen. You wonder whether adjunctive corticosteroids improve outcomes.

P	
I	
C	
O	
Clinical Question	

From PICO to Search Strategy

Using Vignette 1 (AF & DOACs), here is how to convert a PICO into a PubMed search:

PICO	Key Terms	MeSH Terms
P – Adults with AF	atrial fibrillation	"Atrial Fibrillation"[MeSH]
I – DOACs	DOAC, apixaban, rivaroxaban, dabigatran, edoxaban	"Administration, Oral"[MeSH] AND "Anticoagulants"[MeSH]
C – Warfarin	warfarin, coumadin	"Warfarin"[MeSH]
O – Stroke, bleeding	stroke prevention, major bleeding	"Stroke"[MeSH] OR "Hemorrhage"[MeSH]

Boolean combination:

("Atrial Fibrillation"[MeSH]) AND ("Anticoagulants"[MeSH] OR apixaban OR rivaroxaban OR dabigatran) AND ("Warfarin"[MeSH]) AND ("Stroke"[MeSH] OR "Hemorrhage"[MeSH])

Bad PICO vs Good PICO

Bad PICO

"Is aspirin good for the heart?"

Good PICO

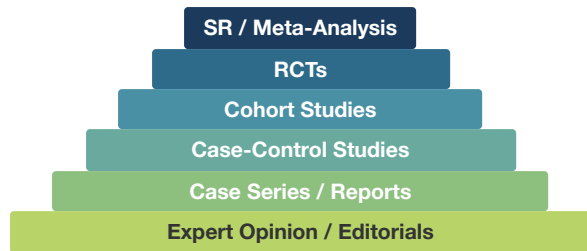
"In adults >50 with no prior CVD (*P*), does low-dose aspirin

Too vague — no defined population, comparison, or measurable outcome.

(I) compared with placebo *(C)* reduce the incidence of first MI at 5 years *(O)*?”

Evidence Pyramid & Levels / Grades Quick Reference

Evidence Pyramid



Levels of Evidence

Level	Description
I	SR of RCTs or single large RCT with definitive results
II	Smaller RCTs with less definitive results
III	Cohort and case-control studies
IV	Case series, case reports
V	Expert opinion without critical appraisal

The Courtroom Analogy

EBM Concept	Courtroom Equivalent
Clinical question	The charge
Study evidence	Witness testimony
Level of evidence	Credibility of witness
Grade of recommendation	Verdict strength
Systematic review	All testimony combined

Grades of Recommendation

Grade	Meaning
A	Strong evidence (consistent Level I studies)
B	Moderate evidence (consistent Level II/III)
C	Limited evidence (Level IV/V)
D	Conflicting or inconclusive evidence

GRADE Certainty of Evidence

Certainty	Interpretation	Symbol
High	Very confident the true effect lies close to the estimate	●●●●
Moderate	True effect is likely close but may be substantially different	●●●○
Low	True effect may be substantially different from the estimate	●●○○
Very Low	Very little confidence; true effect likely substantially different	●○○○

Factors that lower certainty: risk of bias, inconsistency, indirectness, imprecision, publication bias. Factors that raise: large effect, dose-response, confounding would reduce effect.

Oxford Centre for Evidence-Based Medicine — Levels by Question Type

Level	Therapy / Prevention	Diagnosis	Prognosis
1	SR of RCTs	SR of cross-sectional studies with consistently applied reference standard	SR of inception cohort studies
2	Individual RCT or observational study with dramatic effect	Individual cross-sectional study with good reference standard	Inception cohort study
3	Non-randomized controlled cohort / follow-up study	Non-consecutive study or without consistently applied reference standard	Cohort study or control arm of RCT
4	Case series, case-control studies	Case-control study, poor or non-independent reference standard	Case series or case-control, or poor quality prognostic cohort
5	Mechanism-based reasoning or expert opinion		

OCEBM Levels of Evidence Working Group. "The Oxford Levels of Evidence 2." Oxford Centre for Evidence-Based Medicine. <https://www.cebm.ox.ac.uk/resources/levels-of-evidence>

“Name That Study Design” Exercise

Read each study description below and identify the study design. Write your answer on the line provided.

1. “We enrolled 10,000 nurses and followed them for 20 years, recording their diet and tracking cancer incidence.”

Study design: _____

2. “We identified 200 patients with lung cancer and 200 matched controls without lung cancer, then compared their prior smoking history.”

Study design: _____

3. “We randomly assigned 500 patients with hypertension to Drug A or placebo and measured blood pressure outcomes at 6 months.”

Study design: _____

4. “We surveyed 1,000 adults in the community about their blood pressure readings and salt intake on a single day.”

Study design: _____

5. “We pooled data from 15 randomized controlled trials on statins for primary cardiovascular prevention and calculated a pooled effect estimate.”

Study design: _____

Quick Reference: Study Design Identification Tips

Design	Direction	Key Feature	Best For
RCT	Forward	Random allocation to intervention	Therapy
Cohort	Forward or Backward	Follow exposed vs unexposed over time (prospective or retrospective)	Prognosis, Harm, Etiology
Case-Control	Backward	Start with outcome, look back at exposure	Rare diseases, Harm
Cross-Sectional	Snapshot	Exposure & outcome at same time point	Prevalence
SR / Meta-Analysis	Synthesis	Pool multiple studies quantitatively	Summary estimate
Case Series	Descriptive	Series of patients with outcome, no comparator	Novel findings

Tip: Ask yourself three questions: (1) Was there an intervention? (2) Was there random allocation? (3) What is the direction of inquiry (forward, backward, or snapshot)?

Reporting Checklists: CONSORT & STROBE

Use these abbreviated checklists during journal club and when appraising studies. Check off each item as you evaluate a paper.

CONSORT — for RCTs

Consolidated Standards of Reporting Trials

	Item
☐	1. Title & Abstract — Identified as RCT in title
☐	2. Eligibility Criteria — Inclusion/exclusion clearly stated
☐	3. Interventions — Details of each arm described
☐	4. Outcomes — Pre-specified primary & secondary outcomes
☐	5. Randomization — Sequence generation & allocation concealment
☐	6. Blinding — Who was blinded (patients, providers, assessors)
☐	7. Sample Size — How determined (power calculation)
☐	8. Flow Diagram — Participants through each stage
☐	9. ITT Analysis — Analyzed in assigned groups
☐	10. Harms — Adverse events reported per group

STROBE — for Observational

Strengthening the Reporting of Observational Studies

	Item
☐	1. Study Design — Named in title or abstract
☐	2. Setting — Location, dates, follow-up period
☐	3. Participants — Eligibility, sources, selection methods
☐	4. Variables — Outcomes, exposures, confounders defined
☐	5. Data Sources — Measurement methods described
☐	6. Bias — Efforts to address potential bias
☐	7. Study Size — How size was determined
☐	8. Missing Data — How handled
☐	9. Confounding — Control methods (matching, adjustment)
☐	10. Generalizability — External validity discussed

CONSORT vs STROBE — Key Differences

Feature	CONSORT (RCTs)	STROBE (Observational)
Purpose	Report interventional trials	Report cohort, case-control, cross-sectional studies
Randomization	Required: describe method	Not applicable
Blinding	Describe who was blinded	Not applicable
Confounders	Managed by randomization	Must explicitly address confounding
Flow diagram	CONSORT flow diagram required	Participant flow recommended
Bias focus	Selection, performance, detection, attrition	Selection, information, confounding
Full checklist	25 items (www.consort-statement.org)	22 items (www.strobe-statement.org)

Note: These checklists are also your tools for Journal Club presentations. Use them to structure your critical appraisal systematically.

BRIDGE Trial — Critical Appraisal Worksheet

Trial Summary: BRIDGE

Citation: Douketis JD et al. *N Engl J Med* 2015;373:823–833. DOI: 10.1056/NEJMoa1501035

Design: Multicentre, double-blind, randomised, placebo-controlled trial (108 sites, US & Canada)

Population: 1,884 patients with atrial fibrillation on warfarin requiring interruption for elective procedures

Intervention: No bridging (placebo) vs bridging with LMWH (dalteparin 100 IU/kg BID)

Primary Outcomes: Arterial thromboembolism (noninferiority) & major bleeding (superiority)

PART A — VALIDITY (INTERNAL)

Q1. Was the assignment of patients randomised? Was allocation concealment adequate? How was it done?

Q2. Were patients, clinicians, and outcome assessors blinded? How was blinding maintained?

Q3. Were groups similar at baseline? What key variables should be checked?

Q4. Was follow-up complete? What percentage was lost? Was loss balanced between groups?

Q5. Were patients analysed in the groups to which they were randomised (intention-to-treat)?

PART B — RESULTS

Q6. What were the primary outcomes? Report the effect sizes and 95% CIs. Was noninferiority met? Was superiority demonstrated?

Q7. For major bleeding: Calculate the ARR, RRR, and NNT. What do these numbers mean clinically?

PART C — APPLICABILITY

Q8. Are these patients similar to yours? Consider: type of AF, CHADS score, type of procedures, your clinical setting.

Q9. Based on your appraisal, would you forgo bridging anticoagulation in your next patient with AF needing an elective procedure? Why or why not?

CONSORT Quick Reference for This Trial

- ☐ Randomisation described ☐ Allocation concealment ☐ Blinding (double) ☐ Sample size justified ☐ Flow diagram present ☐ ITT analysis
- ☐ Primary outcome pre-specified ☐ Harms reported

BRIDGE Trial — Reference Summary

For use during the Results & Measures discussion

BRIDGE: Bridging Anticoagulation in Patients who Require Temporary Interruption of Warfarin Therapy for an Elective Invasive Procedure or Surgery

Authors: Douketis JD, Spyropoulos AC, Kaatz S, et al.

Journal: New England Journal of Medicine, 2015; 373:823–833

DOI: 10.1056/NEJMoa1501035

Study Design & Population

Feature	Details
Design	Double-blind, randomized, placebo-controlled noninferiority trial
Population	1,884 patients with atrial fibrillation on warfarin therapy
Setting	108 sites in the US and Canada
Clinical context	Warfarin interrupted for elective invasive procedures or surgery
Intervention	No bridging anticoagulation (placebo)
Comparator	Bridging with LMWH (dalteparin 100 IU/kg twice daily)

Key Results

Outcome	No Bridging	Bridging (LMWH)	Measure
Arterial thromboembolism (primary efficacy)	0.4%	0.3%	Noninferiority met ($p = 0.01$ for noninferiority)
Major bleeding (primary safety)	1.3%	3.2%	RR 0.41 (95% CI 0.20–0.78), $p = 0.005$

Worked Example: Calculating Key Measures

Absolute Risk Reduction (ARR):

$$ARR = CER - EER = 3.2\% - 1.3\% = 1.9\%$$

Number Needed to Treat (NNT):

$$NNT = 1 / ARR = 1 / 0.019 = 53$$

For every 53 patients managed without bridging, 1 major bleed is prevented.

Relative Risk Reduction (RRR):

$$RRR = ARR / CER = 1.9\% / 3.2\% = 59\%$$

Relative Risk (RR):

$$RR = EER / CER = 1.3\% / 3.2\% = 0.41$$

No-bridging reduced risk of major bleeding by 59% relative to bridging.

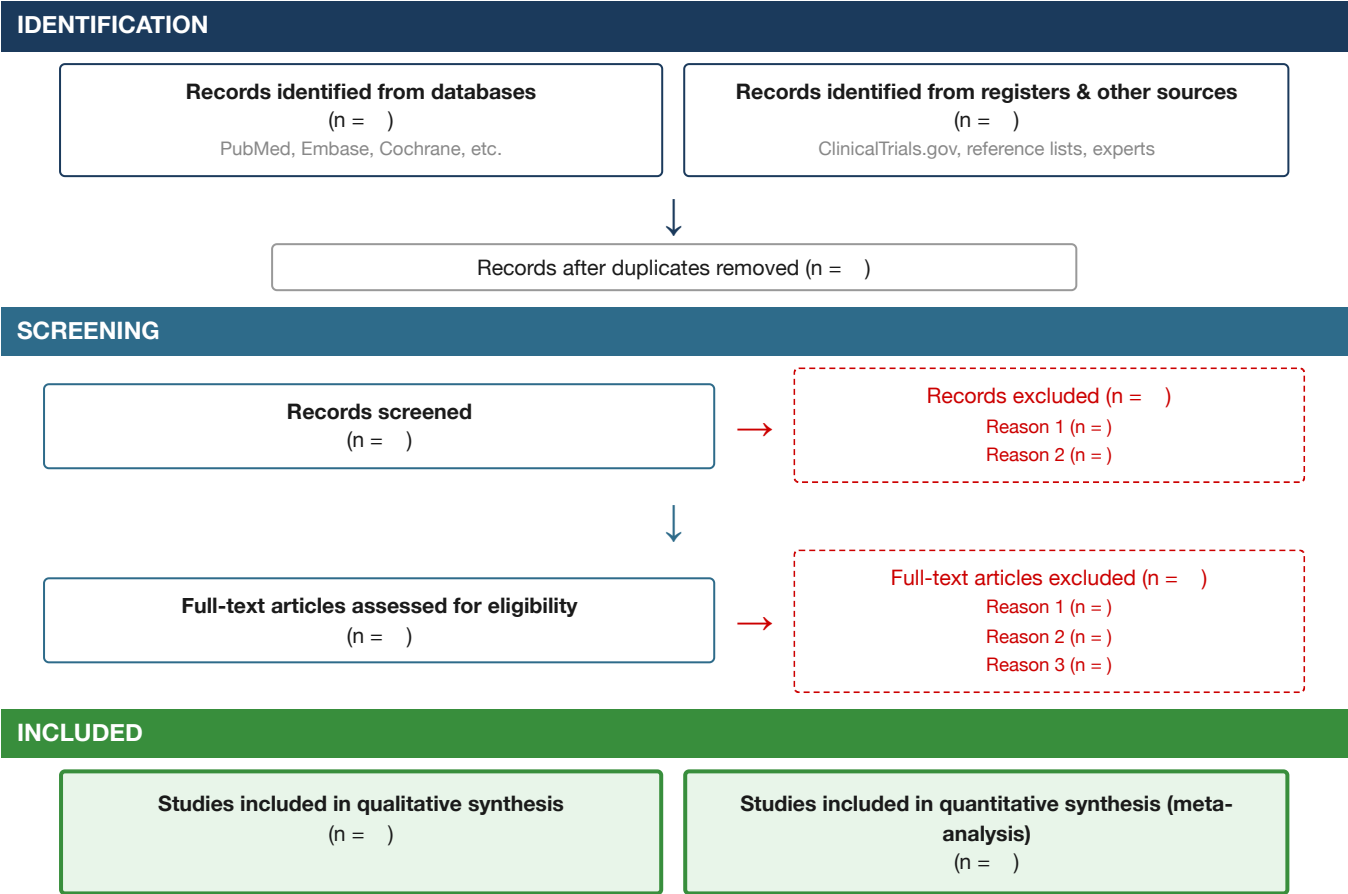
Key Formulas

Measure	Formula	Interpretation
ARR	$ CER - EER $	Absolute difference in event rates
RRR	ARR / CER	Proportional reduction in risk
RR	EER / CER	<1 favors intervention; >1 favors control
OR	$(a/b) / (c/d)$ from 2×2 table	Approximates RR when events are rare
NNT	$1 / ARR$	Patients treated per 1 event prevented
NNH	$1 / ARI$	Patients treated per 1 additional harm

Clinical Bottom Line: In AF patients requiring temporary warfarin interruption for elective procedures, forgoing bridging anticoagulation was noninferior for preventing thromboembolism and resulted in significantly less major bleeding.

PRISMA 2020 Flow Diagram

Template for reporting systematic review study selection. Complete during your SR appraisal.



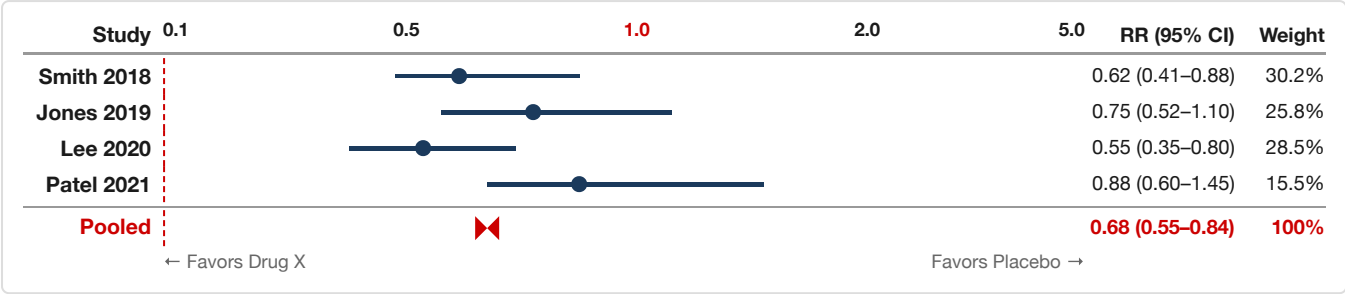
Key Updates from PRISMA 2009 to PRISMA 2020

Change	Details
Databases vs Registers	Sources now separated: databases (PubMed, Embase) vs registers (ClinicalTrials.gov) vs other sources (reference lists, experts)
Automation tools	Report use of automation tools for deduplication, screening, or data extraction
Exclusion reasons at each stage	Reasons for exclusion must be reported at both title/abstract screening and full-text stages
Previous vs new studies	For updated reviews, distinguish previously included studies from newly identified ones
Expanded checklist	27-item checklist with expanded abstract checklist

Citation: Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71

Forest Plot Reading Guide

Sample Forest Plot: Drug X vs Placebo for Mortality



Anatomy of a Forest Plot

Element	Meaning
● Point estimate	Effect size from each study (square size = weight)
— Horizontal line	95% confidence interval for each study
◇ Diamond	Pooled effect estimate; width = pooled CI
Vertical line	Line of no effect (RR=1, OR=1, or MD=0)
Weight %	Contribution of each study to pooled estimate

I² Heterogeneity Interpretation



5 Questions When Reading a Forest Plot

- 1. **Direction:** Does the pooled estimate favor intervention or control?
- 2. **Significance:** Does the diamond cross the line of no effect?
- 3. **Precision:** How wide is the confidence interval?
- 4. **Consistency:** Do individual studies agree? (check I²)
- 5. **Magnitude:** Is the effect clinically meaningful?

Funnel Plot & Publication Bias

A funnel plot graphs each study's effect size (x-axis) against its precision/sample size (y-axis).

- **Symmetric funnel** = no publication bias suspected
- **Asymmetric funnel** (missing lower-left) = small negative studies may be missing
- Egger's test provides a statistical assessment

Key Rule: If the 95% CI of the pooled diamond crosses the line of no effect, the result is *not* statistically significant at the 5% level — regardless of how many individual studies show benefit.

PubMed Search Cheat Sheet

Boolean Operators

Operator	Function	Example
AND	Narrows (both terms)	diabetes AND metformin
OR	Broadens (either term)	heart attack OR myocardial infarction
NOT	Excludes term	asthma NOT pediatric

Operators must be CAPITALIZED. Use parentheses to group: (A OR B) AND C

MeSH Terms

Medical Subject Headings are PubMed's controlled vocabulary.

- **Finding MeSH:** Search → MeSH Database (in PubMed sidebar)
- **Syntax:** "Atrial Fibrillation"[MeSH]
- **Explosion:** Automatically includes narrower terms
- **No explosion:** "Atrial Fibrillation"[MeSH:NoExp]

Useful Filters

Filter	What It Does
Humans	Exclude animal studies
English	Language filter
Clinical Trial	RCTs and clinical trials
Systematic Review	SRs and meta-analyses
Free Full Text	Open access articles only
Date range	e.g., last 5 years

Search Strategy Template (PICO-Based)

Step	PICO	Action
1	P	Identify MeSH & free-text terms for population
2	I	Identify MeSH & free-text terms for intervention
3	C	Identify MeSH & free-text terms for comparator (if needed)
4	O	Consider adding outcome terms (often too restrictive — use cautiously)
5	—	Combine within each concept with OR , then combine concepts with AND
6	—	Apply filters (article type, date, language, humans)

Worked Example: AF & DOACs

```
#1 "Atrial Fibrillation"[MeSH] OR "atrial fibrillation"[tiab]
#2 "Anticoagulants"[MeSH] OR apixaban[tiab] OR rivaroxaban[tiab] OR dabigatran[tiab] OR
edoxaban[tiab] OR DOAC[tiab] OR NOAC[tiab]
#3 "Warfarin"[MeSH] OR warfarin[tiab] OR coumadin[tiab]
#4 #1 AND #2 AND #3
#5 #4 AND (randomized controlled trial[pt] OR systematic review[pt]) AND humans[MeSH] AND
2015:2026[dp]
```

Where to Start Your Search

Source	Best For
Cochrane Library	Systematic reviews
PubMed / MEDLINE	Primary studies, broad search
UpToDate	Quick clinical answers
NICE / WHO Guidelines	Practice recommendations
ClinicalTrials.gov	Ongoing/unpublished trials

Common Search Tags

Tag	Searches	Example
[MeSH]	MeSH heading	"Diabetes Mellitus"[MeSH]
[tiab]	Title/Abstract	metformin[tiab]
[ti]	Title only	SGLT2[ti]
[au]	Author	Douketis JD[au]
[pt]	Publication type	randomized controlled trial[pt]
[dp]	Date of publication	2020:2025[dp]

Pro Tips

- Use **quotation marks** for exact phrases: "heart failure"
- Use **asterisk** * for truncation: therap* finds therapy, therapies, therapeutic
- Start broad, then narrow with filters
- Save your search: "Create alert" for email updates
- Use Clinical Queries for pre-built therapy/diagnosis/prognosis filters

Evidence Hierarchy for Searching

1. Clinical practice guidelines
2. Systematic reviews / meta-analyses
3. Critically appraised topics (CATs)
4. Individual RCTs
5. Observational studies
6. Background information / textbooks

Start from the top and work down. The higher you find your answer, the less appraisal work is needed.